

# Thorpe Desktop — Problems Solved

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**Prepared for:** Patent counsel

**Product:** Thorpe Desktop

**Core invention:** Jonathan — autonomous AI technician with local repair orchestration

**Date:** June 27, 2026

**Software version:** 1.1.0

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## Executive summary

Thorpe Desktop is a cross-platform desktop application (Windows, macOS, Linux) that embeds **Jonathan**, an autonomous AI technician. Jonathan does not merely advise users—it **diagnoses, executes, and verifies repairs locally** on the end-user device, then explains what was done in plain language.

The core technical problem solved: **conventional IT support chatbots recommend manual steps they never perform, cannot verify, and cannot safely execute on the client device.** Thorpe couples a governed, offline-capable local repair engine with an AI that only narrates repairs already executed and recorded.

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## Field of the invention

Computer systems diagnostics, automated remediation, and AI-assisted technical support on end-user desktops and laptops across Windows, macOS, and Linux.

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## Problems in the prior art

### 1. Chat-only support (no local execution)

Traditional IT chatbots and virtual assistants tell users to open Settings, run terminal commands, or reboot. They do not execute OS-level diagnostics or repairs on the device. The user must interpret and perform every step manually.

### 2. Cloud dependency

Most AI support tools require continuous cloud connectivity. When the network is down—the very condition users often need help with—cloud AI cannot assist with local diagnostics or remediation.

### 3. No verification of outcomes

Existing tools do not prove that a suggested or attempted fix improved system health. There is no structured before/after health verification tied to the support interaction.

4. Conflation of AI text with system changes

In conventional LLM-based support, the model may describe actions it did not perform. Users cannot distinguish between a recommendation and an executed repair. This creates trust, liability, and safety risks.

5. Unsafe autonomous agents

General-purpose LLM tool-use agents can invoke arbitrary or hallucinated commands. There is no cryptographic allowlist governing which OS-level actions an AI may run on an endpoint.

6. Reactive-only support

Most consumer and SMB support is user-initiated. Proactive monitoring (CPU, memory, disk) is typically siloed in MSP/RMM tools—not integrated into the same conversational repair pipeline the end user interacts with.

7. Fragmented network troubleshooting

Network problems are often addressed with disconnected utilities or generic advice. There is no unified, offline-capable connectivity diagnostic suite fused into a single agent incident response.

8. Lack of enterprise governance

Consumer AI assistants lack role-based controls, repair-pack signing, audit trails, and policy-governed separation between planning AI and narration AI.

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How Thorpe Desktop solves these problems

| Problem                     | Thorpe / Jonathan solution                                                                                                   |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Manual chat-only support    | Executes local diagnostics and repairs (connectivity suite, DNS flush, temp cleanup, resource analysis, disk analysis, etc.) |
| Cloud dependency            | Core diagnostics, repair execution, and local narrative operate <b>offline</b> ; cloud LLM is optional                       |
| No fix verification         | After mutating repairs, runs a <b>follow-up health scan</b> and reports before/after metrics in the structured response      |
| AI invents fixes in text    | <b>Execute-then-narrate</b> : repairs complete <b>before</b> narrative generation; AI constrained to executed repairs only   |
| Unsafe autonomous execution | <b>Mutating</b> actions require explicit user approval; safe diagnostics run automatically                                   |

|                              |                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------|
| Uncontrolled tool invocation | Repair actions filtered by <b>cryptographically signed repair-pack allowlist</b>                          |
| Reactive-only support        | <b>Proactive Watchdog</b> monitors CPU, memory, and disk; alerts hand off into the same incident pipeline |
| Fragmented troubleshooting   | <b>Offline connectivity suite</b> fuses layered network diagnostics into one agent response               |
| No audit trail               | Collects system evidence, persists incident sessions, optional PSA/webhook escalation                     |

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Technical architecture (summary)

Jonathan implements a local-first incident pipeline:

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User message → Evidence collection → Security gate → Offline connectivity (if applicable)
→ Repair planning (LLM JSON or rules) → Repair-pack allowlist filter → Confirmation gate
→ Local OS repair execution → Post-mutating verification scan → Session persistence
→ Narrative generation (local template or cloud LLM) → Structured chat response
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**Key invariant:** Cloud LLM failure does not prevent repairs from having already executed locally.

**Primary code reference:** orchestrate\_incident in src-tauri/src/agent/mod.rs

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Commercial and user problems solved

**End users without IT staff** — Natural-language fix for “Wi-Fi not working” or “computer is slow” without a help desk or MSP.

**Small businesses** — Reduces paid IT visits for routine maintenance and provides documented, verifiable remediation.

**MSP / enterprise (adjacent)** — Optional PSA webhook escalation; governed AI with feature and policy controls rather than unconstrained autonomous agents.

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Recommended patent focus

| Priority | Application                   | Description                                                           |
|----------|-------------------------------|-----------------------------------------------------------------------|
| P0       | Patent A — Core orchestration | Execute-then-narrate autonomous IT incident pipeline on client device |

|           |                                 |                                                                    |
|-----------|---------------------------------|--------------------------------------------------------------------|
| <b>P0</b> | Patent B — Repair packs         | Cryptographically signed allowlist gates which OS tools AI may run |
| <b>P0</b> | Patent C — Offline connectivity | Layered network diagnostics fused into agent; no cloud required    |
| <b>P1</b> | Patent D — Watchdog handoff     | Background monitor hands alerts into same agent pipeline           |
| <b>P1</b> | Patent E — Dual LLM             | Separate planner and narrator LLMs with independent fallbacks      |

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One-sentence invention statement

Thorpe Desktop solves the problem that AI IT assistants recommend fixes they never perform, cannot verify, and cannot safely execute locally—by coupling a governed, offline-capable repair engine on the client device with an AI that only narrates repairs already executed and verified.

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Related documents in this package

- 01-INVENTION-DISCLOSURE-JONATHAN.md — Full technical disclosure
- 02-PATENT-CANDIDATE-INVENTORY.md — Patent applications A–H
- 04-CLAIM-SEEDS-FOR-ATTORNEY.md — Seed claims for drafting
- 05-PRIOR-ART-DIFFERENTIATION.md — Prior-art categories and differentiation
- 07-EXECUTIVE-BRIEF-ONE-PAGE.md — One-page summary

**Applicant / assignee:** [Complete Document 06 before filing]